

Problem:

A single (15×15×3) image is passed through a convolutional layer with 28 filters, each of size (3×3×3). The padding size is 1 (1 unit at top, bottom, left, and right) and the stride size is also 1.

Solution:

1. Output Feature Map Size: 15×15×28

Explanation: Using the formula $(N\text{-size} - F\text{-size} + 2 \times (\text{Padding})) / (\text{Stride}) + 1$ for both height and width:

Height: $(15 - 3 + 2 \times 1) / 1 + 1 = 15$

Width: $(15 - 3 + 2 \times 1) / 1 + 1 = 15$

Depth: Number of filters = 28

Therefore, output volume is 15×15×28

2. Number of Parameters:

Parameters per filter: (3×3×3) weights + 1 bias = 28 weights + 1 bias = 29

Total parameters: 29 × 28 filters = 784

This can also be calculated as: $(3 \times 3 \times 3 \times 28) + 28 = 756 + 28 = 784$